



## FOREST VERIFICATION REPORT

### ANALYZED ZONE INFORMATION

|                |         |               |            |
|----------------|---------|---------------|------------|
| Latitude       | 39.8837 | Longitude     | -88.1543   |
| Zone size (km) | 10      | Analysis date | 2026-08-27 |

47

FOREST INTEGRITY SCORE / 100 Degraded Forest

### DETAILED RESULTS

| Indicator                         | Value                                                                                            | Interpretation                   |
|-----------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------|
| Current vegetation (NDVI)         | 0.428                                                                                            | Good                             |
| NDVI Trend 3 years (Sentinel-2)   | +0.007                                                                                           | Stable                           |
| 10-year trend (Landsat 8/9)       | +0.209                                                                                           | Improving                        |
| 40-year trend (Landsat 1984)      | +0.242                                                                                           | Improving                        |
| Forest Status (FAO)               | Forest recovery                                                                                  | Recuperation                     |
| Annual NDVI rate                  | +0.6% / year                                                                                     | Amelioration                     |
| NDVI change 5 years               | -0.025                                                                                           | Stable                           |
| Leaf type                         | Broadleaf (Deciduous/Evergreen)                                                                  | Flat leaves, high photosynthesis |
| Developmental stage               | Growing forest (20-60 years)                                                                     | Active growth phase              |
| Above-ground biomass              | ~84.5 t/ha                                                                                       | Medium biomass (50-150 t/ha)     |
| Canopy height                     | ~13.7 m                                                                                          | Estimated from NDVI              |
| AI Forest Type                    | Savanna/Woodland                                                                                 | Open tree cover, dry season      |
| Sentinel-1 Radar (VV)             | -11.124 dB                                                                                       | Moderate forest (radar)          |
| Forest cover                      | 56.6%                                                                                            | Low                              |
| Fire pixels detected (FIRMS/year) | 232 pixels                                                                                       | ~58 events estimated             |
| Overall trend                     | stable                                                                                           |                                  |
| Satellites used                   | - ESA Sentinel-2<br>- NASA FIRMS<br>- ESA Sentinel-1<br>- NASA Landsat 8/9<br>- NASA Landsat 4/5 |                                  |
| Deforestation Alert               | POSITIVE — Forest recovery detected                                                              |                                  |
| Area change (3 years)             | +327.3 ha                                                                                        | +4.66%                           |
| Annual rate (area/yr)             | +109.1 ha/yr                                                                                     | Recovery                         |
| Area change (10 years)            | +7347.3 ha                                                                                       | +104961.43%                      |

|                         |              |          |
|-------------------------|--------------|----------|
| Annual rate 10yr (area) | +734.7 ha/yr | Recovery |
|-------------------------|--------------|----------|

SCORE BREAKDOWN

| Component                          | Points | Maximum |
|------------------------------------|--------|---------|
| Current vegetation (NDVI)          | 11     | 25      |
| NDVI trend (3 years)               | 15     | 25      |
| Forest cover                       | 11     | 25      |
| Sentinel-1 Radar (VV)              | 10     | 15      |
| Fire score (0=many fires, 10=none) | 0      | 10      |
| TOTAL                              | 47     | 100     |

FIRE DETECTION METHODOLOGY: Fire pixels are detected by NASA FIRMS using VIIRS (375m) and MODIS (1km) thermal anomaly sensors. Raw pixel count represents thermal detections over 12 months in the analysis zone. Estimated fire events are calculated using NASA clustering methodology (~4 pixels per fire event average). This is a statistical approximation. For precise fire event mapping, consult NASA FIRMS directly at [firms.modaps.eosdis.nasa.gov](https://firms.modaps.eosdis.nasa.gov)

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